

The impact of a pharmacist intervention on 6-month outcomes in depressed primary care patients

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Received 31 January 2003; accepted 12 August 2003

Abstract

The object of the study was to evaluate outcomes of a randomized clinical trial (RCT) of a pharmacist intervention for depressed patients in primary care (PC). We report antidepressant (AD) use and depression severity outcomes at 6-months. The RCT was conducted between 1998 and 2000 in 9 eastern Massachusetts PC practices. We studied 533 patients with major depression and/or dysthymia as determined by a screening test done at the time of a routine PC office visit. The majority of participants had recurrent depressive episodes (63.5% with ≥ 4 lifetime episodes), and 49.5% were taking AD medications at enrollment. Consultation in person and by telephone was performed by a clinical pharmacist who assisted the primary care practitioner (PCP) and patient in medication choice, dose, and regimen, in accordance with AHCPR depression guidelines. Six-month AD use rates for intervention patients exceeded controls (57.5% vs. 46.2%, $P = .03$). Furthermore, the intervention was effective in improving AD use rates for patients not on ADs at enrollment (32.3% vs. 10.9%, $P = .001$). The pharmacist intervention proved equally effective in subgroups traditionally considered difficult to treat: those with chronic depression and dysthymia. Patients taking ADs had better modified Beck Depression Inventory (mBDI) outcomes than patients not taking ADs, (-6.3 points change, vs. -2.8 , $P = .01$) but the outcome differences between intervention and control patients were not statistically significant (17.7 BDI points vs. 19.4 BDI points, $P = .16$). Pharmacists significantly improved rates of AD use in PC patients, especially for those not on ADs at enrollment, but outcome differences were too small to be statistically significant. Difficult-to-treat subgroups may benefit from pharmacists' care. © 2004 Elsevier Inc. All rights reserved.

Keywords: Depression; Primary care; Clinical pharmacist; Antidepressant medication; Outcomes

1. Introduction

Despite recent advances in both screening and treatment protocols [1–15], policy makers report that the undertreatment of depression in primary care settings remains a serious public health problem [16–35]. Unfortunately, in cases where depression may be related to the patient's presenting medical symptoms (i.e., fatigue, pain, and/or lack of appetite) [36–38], the primary care physician (PCP) and the

patient may collude in keeping this psychiatric illness hidden. Whereas the patient may choose not to report depressive symptoms, the PCP may lack the time or the expertise to inquire about them [34,39,40]. Other barriers to treating depression in primary care settings include erratic follow-up and the reluctance of some patients to keep taking prescribed antidepressant (AD) medication [3,5,6,41,42]. In addition, progress in identifying and treating depression in primary care settings has been particularly slow for patients with low income and those with chronic recurrent depressive episodes [33,34].

Given that AD medications represent a cost-effective means to treat depression [19,27] we hypothesized that an intervention that increased AD use among depressed patients in primary care settings would enhance outcomes. We

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Funding Source: Grant MH56214 from the National Institute of Mental Health, Rockville, MD.

also hypothesized that extenders to primary care physicians would be able to increase AD use. While prior studies have examined adding psychiatrists [3,5], psychologists [4,13], nurses [1,8,13,43], and case managers [7,11], to the primary care team, we decided to add clinical pharmacists because of their sophisticated knowledge concerning medication management [44–46]. In recent years, pharmacists have received increasingly rigorous clinical training, and we assumed that they could assist PCPs by documenting complete medication histories (thereby preventing problems such as adverse reactions or drug interactions), providing medication counseling, and facilitating communication with patients. Researchers focusing on other illnesses (e.g., such as asthma and heart disease) have relied on a similar pharmacist intervention [44,45,47–50].

We conducted a randomized, controlled trial of a pharmacist intervention in nine different primary care sites in eastern Massachusetts. Numerous other researchers have conducted efficacy studies of interventions geared toward primary care patients with untreated depression [3,4,10,11]. In contrast to these efficacy studies that typically eliminate many categories of depressed patients (e.g., such as those currently on AD medications and those with comorbid anxiety and personality disorders), our study screened consecutive patients visiting these nine sites for a routine office visit for both major depressive disorder (MDD) and dysthymia. We then enrolled all patients who screened positive for depression and/or dysthymia regardless of whether they were currently using AD medications. This is the first RCT in the literature to examine the clinical pharmacist's role in the treatment of depression in primary care.

2. Methods

2.1. Study setting and physicians

The study population was recruited from nine primary care practices in metropolitan Boston: 5 general medical practices at an academic medical center, 1 urban, and 2 suburban community Boston medical practices, and a community health center. Of the 53 primary care practitioners, 45 were internists, 5 were family practitioners, and 3 were nurse practitioners. Care was provided according to the usual primary care model with PCPs and nurses delivering care themselves or referring to mental health specialists. This is typical for the Boston area, which has a high penetration mental health use. All physicians belonged to group practices that participate in independent practice associations (IPAs), and preferred provider organizations (PPOs). Physicians of different ages, educational backgrounds, and styles of treating depression were recruited. All primary care physicians and nurse practitioners (collectively referred to as PCPs) received the AHCPR Depression Guidelines [19] booklet before enrollment of patients, but the study did not provide any clinical training on how to treat depression.

2.2. Screening, recruitment and treatment assignment

There were 16,766 consecutive patients presenting to the study sites for routine care who completed a self-administered health survey, the PC-SAD© (Primary Care Screener for Affective Disorders) [51] that was administered by a research assistant stationed in the PCP's office. This paper-and-pencil instrument was scored by a computer later the same day. The PC-SAD provides a DSM-IV symptom count and DSM-IV diagnosis for major depressive disorder (MDD) and dysthymia. By the end of the day of the visit, the PCPs received a full report on all patients screening positive, regardless of whether they were eligible for the study. The PCP was asked to complete a brief patient history that included any contraindications against participation in the study (e.g., such as severe congestive heart failure with multiple systems disease) and to confirm the diagnosis.

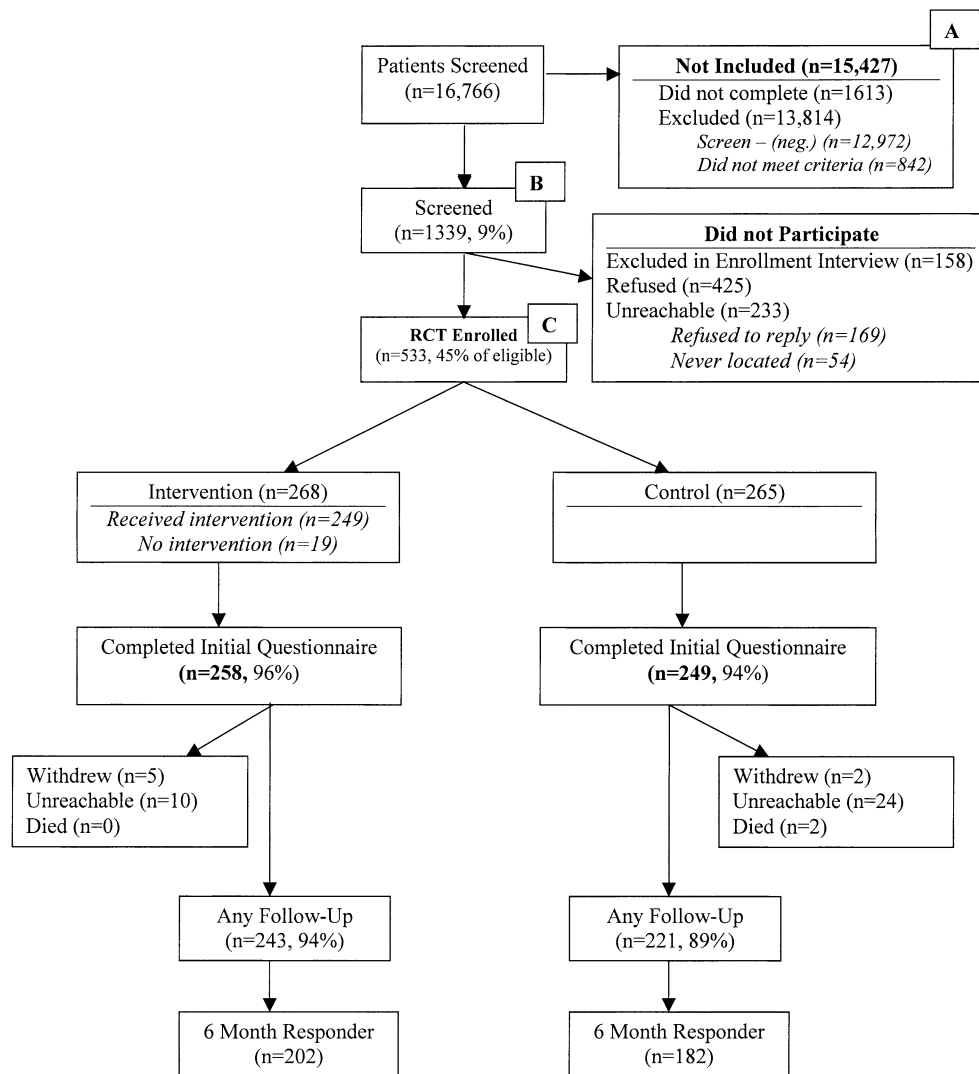
Patients were eligible if they: 1) received care from a PCP in any site; 2) met DSM-IV criteria for major depressive disorder (MDD) and/or dysthymia; 3) were 18 years of age or older; 4) could read and understand English; 5) had no acute life threatening condition with a terminal prognosis of <6 months; 6) were not pregnant (or had not given birth within the last 6 months). Like numerous other researchers [1,3,5,7,8,10,11,13,52], we excluded patients with current alcoholism (defined as more than one positive response on the CAGE, plus one item assessing current usage), bipolar disorder, and/or psychotic disorders. But patients with lifetime alcoholism, long-term/chronic depression (those with ≥ 4 MDD episodes in their lifetime plus their first diagnosis > 10 years ago), anxiety disorders, likely personality disorders (as indicated by NEO scores ≥ 17), [5,53], or comorbid medical conditions were not excluded.

Before randomization eligible patients were mailed consent forms (approved by the Institutional Review Board for each site) within a week of their office visit. The documents explained the study and included an invitation from their PCP to participate. Clinician members of the research team called patients within 2 weeks to review the study and invite them to participate. Patients who gave consent were randomized via a 'computerized coin-flip' built into the screener.

2.3. Enrollment and follow-up

Although 16,766 patients agreed to be screened (Fig. 1), 10% (1613) (Fig 1A) failed to complete the screening. Of those who did not complete the screening, the majority (1308/1613, 81%) changed their minds after starting and the remaining (305/1613, 19%) encountered logistical problems (e.g., some patients were called into their appointment before finishing.) In addition, 13,814 patients were not included because they screened negative or did not meet inclusion criteria.

Therefore, of the 15,153 patients who completed the



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Fig. 1. Participant flow.

screening, 1339 (9%, Fig 1B) were eligible for the enrollment telephone call. Patients were classified as suffering from MDD; dysthymia or double depression (DD) (if they met criteria for both diagnoses). Of these 1339 patients, 158 were excluded, 425 refused, and 223 were unreachable. 533 (45%) (Fig 1C) gave consent. There were no significant differences between the enrolled and nonenrolled in terms of age, gender distribution, generic mental health functioning [as assessed by the mental health index of the SF-36 (MHI)], or physical functioning [assessed by the physical component summary score (PCS)], derived from SF-36 items [54,55]. Nonenrollees reported fewer depression symptoms (5.2 vs. 6.5, $P < .01$). Among a sub-sample of screened patients, an insignificantly larger fraction of enrollees indicated a willingness to take ADs if their PCPs suggested it (59% vs. 51%, $P > .2$). More control patients dropped out or were lost to follow-up at 6 months (26 vs. 15, $P = .07$). Two controls died of medical complications. There were 7% (19/268) of the patients in the intervention

group who dropped out without ever participating in the study.

Of the 533 enrolled RCT patients, 507 successfully completed the initial questionnaire, including 44 by telephone. Initial and follow-up questionnaires were obtained independent of random assignment. The number of respondents listed in the results tables vary because of study participation status and missing data on specific questions.

2.4. Pharmacist intervention

Patients were randomized to either the intervention arm or control arm (which consisted of standard PCP care), regardless of whether they were taking AD medications. The intervention was guided by the use of a protocol based on clinical pharmacy principles and AHCPR guidelines [19], and did not involve prescribing a specific AD medication. The protocol emphasized: 1) obtaining a thorough medication history [56], 2) assessing a patient's medication

regimen for drug-related problems (such as side effects or drug interactions), 3) monitoring drug efficacy and toxicity, especially for the symptoms of depression, 4) educating patients about depression and antidepressants, 5) encouraging patients to start and maintain AD therapy, and 6) facilitating communication with a patient's PCP.

The protocol specified that pharmacists contact each patient a minimum of 9 times over 18 months (at 2, 4, 6, 8, and 12 weeks, also at 6, 9, 12, and 18 months). Five doctoral level (Pharm.D.) clinical pharmacists with experience in academic medical centers were trained to administer this protocol. They were urged to establish a strong therapeutic alliance with patients and to improve medication-taking behavior. Though PCPs invited their patients to participate in the study, they did not introduce the pharmacist.

Pharmacists contacted the patients initially by telephone to set up an appointment. After the initial appointment they informed the patient's PCP and provided the PCP with a thorough medication history (including adherence to prescribed medications and drug-related problems) and whatever recommendations the pharmacist may have suggested to improve the regimen. Subsequent physician contacts with pharmacists were based on the individual physician's preferences. Pharmacists reported information to the PCP using a standard computerized template. The completed template was easily transmitted to PCPs and was oftentimes incorporated into the patient's medical record. Pharmacists' time with the PCPs averaged 15 min per patient over 6 months.

Pharmacists spent about 5 times more time with patients than with PCPs. In actuality, the frequency of patient contact was titrated to the patient's particular needs. At 6 months, patients had a mean of four contacts, with 90% having one or more. Patients initiating AD therapy (or a new medication) received more frequent contact than patients continuing on a previous regimen. Obviously, if the patient no longer wanted to interact with the pharmacist or if it became clear that the PCP would not prescribe or the patient would not accept medications for depression, the pharmacist interaction was reduced. On average pharmacists spent 70 min per patient during the six-month intervention (11 min/month). The approximate cost of the pharmacists' time spent with the patient and their PCP averaged between \$80 to \$100 per patient over 6 months.

In addition to their unique pharmacist activities, pharmacists fulfilled some basic patient needs, such as that of general social support and overcoming system inadequacies (subject of another paper) [57]. This area of support is closely related to case management functions described in other intervention studies [1,3–5,7,8,11,13,43,58]. Pharmacists encouraged and facilitated referrals to the mental health specialty sector but the physician initiated the referral. Pharmacists encouraged communication between patient and PCP to address this request. Based on exit interviews patients expressed high satisfaction with the pharmacist intervention.

The investigators held regular meetings to discuss the

protocol and to insure that it was being implemented uniformly. The Principal Investigator, a senior psychiatrist, provided weekly clinical supervision, if needed, for problematic clinical issues (for example, patients revealing suicidal thoughts, complex family situations coexisting with the case).

2.5. Control sample

The PCPs who saw the control patients received the results of the depression screener indicating a DSM-IV diagnosis of major depressive disorder (MDD) and/or dysthymia. Otherwise, control patients received usual care.

2.6. Study measures

All study outcomes were determined by patient self-report. After the in-office screening, patients were mailed surveys at initial enrollment, 3 months and 6 months. Non-responders were contacted by telephone.

To determine whether a patient with MDD or dysthymia was taking an antidepressant, we asked patients to list all current medications. Patients were classified as on an AD if they were taking a medicine with known AD effects. Patients who skipped this item were asked "Do you take any prescription medicines for depression?" and "Are you now taking a prescription medicine for depression?" Patients who responded "yes" to either question were considered to be taking an AD. To validate this approach, we compared these responses with pharmacy claims data, which was available for a subset of patients (approximately 33%). Positive and negative predictive values were 91% for the patient-report approach versus the pharmacy claims [59]. A separate publication [59] reports excellent agreement between self-report AD medication use and pharmacy claims data, and explains the discrepant cases.

Primary study outcomes were AD use rates at 6 months and changes in severity of depression as assessed by a modification of the Beck Depression Inventory (BDI) [60,61] based on the PC-SAD®. The PC-SAD, like other well-known DSM-IV screeners [12,62], can be re-scored to be a depression severity measure using BDI measurement units, with equal discriminatory power to the original BDI. We call this PC-SAD-derived measure a Modified BDI (mBDI) because it has a correlation of 0.89 with the BDI [63] and many of the questions are similar to BDI items. The mBDI scale, like the BDI, ranges from 0–63, with 0 indicating no depression and 63 indicating severe depression. We also computed scores for the MHI-5 (MHI) from the SF-36 [54,64] generic health instrument at screener, initial questionnaire, 3, and 6 months. We did not explicitly measure medication side effects, but we did measure changes in the Physical and Mental Component Summaries (PCS & MCS) [55] from the SF-12.

Additional data on medical and mental health history were collected on the initial questionnaire; in some cases,

this questionnaire was returned after the patient was randomized. Patients reported on previous episodes of depression (using 5 questions about depression history, and if they had any, they were asked about the number of episodes), age, gender, education, current and prior AD use, and comorbid medical conditions. Because these variables are based on reports as opposed to opinions, we assume they are not affected by the randomization procedure.

We considered patients to have a history of AD use if they reported that they first took an antidepressant more than a year before screening. Use that is more recent might be confused with the current episode.

2.7. Statistical analyses

Using patient level ‘intent to treat’ analysis, we analyzed outcomes (AD use and mBDI) with 6-month questionnaire responses. Six-month questionnaires received 3 or more months late (>9 months) were considered missing for these analyses (and thus, were not considered reflective of 6-month outcome). All 384 patients with any 6-month data were included based on their original assigned treatment, even if they never saw a pharmacist.

We analyzed depression-specific clinical outcomes (mBDI) and other generic health outcomes (MHI-5, PCS, and MCS from the SF-12) using linear regression along with AD use rates using logistic regression techniques. The PC-SAD office screener contained prestudy versions of study variables that were used as a baseline to adjust the analysis. All models were adjusted for age, gender, diagnostic category (MDD, Dysthymia, or DD at the time of screening), both mBDI and MHI, and prior AD use. Covariates were used because of tradition (age and gender), accuracy enhancement (diagnostic category and screener mBDI and MHI-5) and initial differences (past AD use)—these categories are appropriate because the treatment groups were randomized.

When we report changes in the percentage of AD use, we calculated change in use as -1 , 0 , or 1 , depending on whether the patient stopped using ADs, continued to use ADs, or started using ADs. We regressed this change on a dummy variable for experimental treatment and covariates, just as we did in the other analysis described above.

Patients with missing data were excluded from the reported results, but we conducted an extensive analysis of the potential bias introduced by missing data. We compared study arms with respect to: 1) the proportion of patients lost in the RCT, 2) screener scores for patients with missing outcomes, and 3) all available outcomes for patients who were missing primary outcomes. Specifically, we compared AD use and mBDI data culled from questionnaires outside the time limits of our primary outcomes. We also conducted versions of the analysis using outcomes scored from the nearest available data, but these were the same as results reported in terms of both significant findings and trends. There were no significant differences between intervention

and control groups in outcomes or in the proportion of patients with missing data.

We also conducted analysis of the effectiveness of AD medications in terms of mBDI points to confirm what we know from clinical trials, that ADs are effective. For these analyses, we took improvement in outcomes from initial enrollment to 6 months as a dependent variable because the sample was not randomized by AD use, but by pharmacist.

We analyzed the impact of several factors that were thought either to interfere with or enhance the impact of the pharmacist. We analyzed AD use for each of three subgroups—patients with chronic depression (defined as ≥ 4 MDD episodes in their lifetime plus first diagnosis >10 years ago), dysthymia, and likely personality disorders (as indicated by either NEO scores ≥ 17 or in the case of intervention patients, clinical review with the study psychiatrist) [53]. We then compared the odds ratios of AD use between the subgroups.

We also analyzed the impact of an indirect effect of the intervention on 6-month outcomes through improved medication use. We considered AD use at 3 months as representative of continued AD use in the first 6 months [65]. This effect is calculated as the product of improvement in AD use as a function of intervention or usual care, and the impact of that AD use on BDI outcomes, assuming parallel impact across treatment groups.

2.8. Power and sample sizes

The study was powered to find a difference of 1.74 BDI points with 80% power. Because this was a pharmacist intervention rather than a medication trial, we assumed that the treatment would achieve only a portion of the effect of a full-scale trial of ADs. Whereas controlled AD trials use rigorous exclusion criteria and invite a homogeneous sample; we did not. For example, we included patients with a history of and/or current AD use. Estimates based on the best available data suggested that only 30% of patients who started taking ADs at enrollment would still be on ADs after 6 months. Based on these assumptions, we calculated that we would need 529 patients at 6 months to achieve 80% power. We did not achieve this number; largely because we were not able to enroll the percentage of patients we expected (70% expected, 45% enrolled) or afford more screening to compensate.

3. Results

Our primary hypothesis was that the intervention patients would be more likely to be using ADs and thus would have better outcomes. After examining the success of the randomization, we evaluated the results for the full study sample. To gain further insight into the results, we also evaluated several sub-samples, namely, patients who were not initially on ADs and patients traditionally considered diffi-

Table 1
Patient characteristics*

Characteristic	N	RCT Enrolled (N = 507)	Experimental (N = 258)	Control (N = 249)	P-value
Mean Age (SD)	507	42.3 (13.9)	42.9 (13.8)	41.7 (14.0)	0.30
Gender (% female)	507	71.8	69.4	74.3	0.24
Race (% white)	500	72.4	72.1	72.8	0.92
Education (% > = 12 years)	499	85.8	85.7	85.8	1.00
Marital Status (% married)	498	29.7	30.7	28.7	0.70
Employed (% > 20 h at screener)	491	60.9	62.5	59.2	0.52
Household income (% < \$10K)	431	17.6	17.0	18.3	0.80
Household income (% > \$80K)	431	14.2	12.8	15.5	0.49
Prior episodes (% > = 2)	493	81.5	80.7	82.4	0.64
Prior episode (% > = 4)	493	63.5	64.3	62.7	0.78
History of AD Use (first meds >= 1 year ago)	501	50.7	56.1	45.2	0.016
Current AD use at initial timepoint	479	49.5	50.6	48.3	0.65

* Includes dropouts.

cult to treat (e.g., because of the chronicity of the depressive disorder). Finally, we also explored a conceptual model of how the intervention works [66].

3.1. Patient characteristics

There were 507 (of the initial 533) evaluable patients followed in the RCT; 258 intervention, and 249 control. Socio-demographics of the study population are listed in Table 1. The ethnic distribution is similar to national data [67], except for a larger Asian population (6%). The socio-demographic characteristics of the patients were: 42.3 years, mean age; 71.8% female; 72.4% White; 29.7% married; 60.9% employed 20 or more hours per week; and 17.6% mean household income <10 K. Overall, 37.1% of patients had seen a psychiatrist or mental health provider in the last 3 months. There were 40% who met the criteria for MDD, 24% for dysthymia, and 36% for DD. There were no differences in these characteristics in any of the intent to treat analyses. There were no significant differences between the control and intervention groups on any socio-demographic variables. Intervention patients were more likely to have first used ADs more than a year before the initial questionnaire. Consequently, we controlled for prior experience with ADs in all subsequent analyses.

Patients not on ADs at enrollment differed from patients already taking ADs in two important respects. They had more prior experience with ADs (63.9% vs. 35.3%, $P < .001$) and had significantly higher mBDI scores (24.9 vs. 23.2, $P = .014$), indicating more depressive symptomatology.

3.2. Primary outcomes: antidepressant use and depression outcomes

The intervention group had more patients on ADs at 3 and 6 months than the control group (3 months, 60.6% vs. 48.9%, $P = .024$; 6 months, 57.5% vs. 46.2% adjusted, $P = .025$) (see Table 2).

Outcomes (mBDI scores) at 6 months favored the intervention group, but the trend did not reach statistical significance (17.7 for intervention vs. 19.4 for control, adjusted, $P = .16$, based on 384 patients who completed both initial and 6 month questionnaires. See Table 3). Results at 3 months were similar (Table 3). Adjusted results at 6 months for the MHI (data not shown) were similar in direction (51.9 vs. 49.0, $P = .15$) and MCS (40.4 vs. 38.6, $P = .19$), but were not statistically significant. Furthermore, there were no differences in 6-month outcomes for PCS (42.9 in both groups). The 95% confidence intervals for mBDI, MHI, and MCS range from no difference to a moderate difference. For example for the mBDI, the 95% confidence interval of the treatment effect ranges from -4.0 points benefit to $+0.6$ points adverse effect. Twelve and 18-month results (not shown) were consistent with the 6 month data. There was sustained AD effect and outcomes had similar direction favoring the intervention but was not statistically significant.

In the next sections, we examine some subgroups to provide further insight into the results and their interpretation.

3.3. Antidepressant (AD) use rates by initial AD taking status

Rates of AD use at baseline, 3 months, and 6 months by initial medication status (off or on AD) are shown in Table

Table 2
Rates* of Antidepressant for All RCT patients

	N	Intervention	Control	P-value
Initial	479	49.6	49.3	0.92
Initial w/o telephone	461	49.3	49.2	0.95
3 month	365	60.6	48.9	0.024
6 month	378	57.5	46.2	0.025

* 3- and 6-month results adjusted for screener covariates: age, gender, mBDI, MHI, and prior AD use at enrollment.

Table 3
Depression outcomes (mBDI*)[†] for All RCT patients

	N	Intervention	Control	P-value
Screeners	512	23.2	23.2	0.97
3 month	370	18.7	19.6	0.35
6 month	384	17.7	19.4	0.16

* Modified Beck Depression Inventory.

[†] 3- and 6-month results adjusted for screener covariates: age, gender, mBDI, MHI, and prior AD use.

4. For patients not on ADs at study entry ($n = 234$), rates of AD use were higher in the intervention group at both 3 months (29.2% vs. 11.0%, $P = .005$) and 6 months (32.3% vs. 10.9% adjusted $P = .001$). For patients using ADs at study entry ($n = 227$), there were no significant differences in AD use between intervention and control groups either at 3 (90.7% vs. 87.2, $P = .50$) or 6 months (83.4% vs. 78.4%, $P = .33$). Note that the 227 (approximately 50% of the sample) patients using ADs at study entry screened positive for major depression, dysthymia, or both, despite ongoing medication use.

Patients not on ADs ($n = 234$) can be further subdivided into the 60.3% who had never taken ADs and the 39.7% who had previously used them but were not doing so when our study began. Of the 60.3% who reported never having used ADs, rates of AD use at 6 months were higher in the intervention group than in the control group (27.1% vs. 5.9%, $P = .008$). Similarly, differences were also found in the 39.7% who had previously used ADs (42.4% vs. 15.1%, $P = .024$).

3.4. Mental health outcomes for patients not on ADs at enrollment

For patients not on ADs at enrollment, mental health outcomes for the intervention patients were no different than control patients, including mBDI (18.1 vs. 19.9, $P = .32$; see Table 5), MHI-5 (52.7 vs. 47.3, $P = .08$), and MCS (41.9 vs. 38.4, $P = .08$) scores. The trend is similar to the outcomes for the full sample (Table 3).

3.5. Rates of AD use in traditionally difficult-to-treat subgroups not on ADs at enrollment

Rates of AD use at 6 months were higher in intervention than control patients who had chronic depression (42.7% vs.

Table 5
Depression outcomes (mBDI*)[†] for patients not on ADs at Enrollment

	N	Intervention (N = 115)	Control (N = 119)	P-value
Screeners	223	22.0	23.0	0.35
3-month	182	18.6	19.3	0.61
6-month	179	18.1	19.9	0.32

* Modified Beck Depression Inventory.

[†] 3- and 6-month results adjusted for screener covariates: age, gender, mBDI, MHI, and prior AD use.

13.9%, $P = .05$, Table 6), dysthymia (47.8% vs. 15.6%, $P = .06$), and potential personality disorder (37.1% vs. 13.4%, $P = .01$). The rates of starting ADs among these traditionally difficult-to-treat subgroups are similar to the rates for all patients not on ADs at enrollment (Table 4, left side).

While gains in AD use over the 6-month period in all study patients were significantly greater in the intervention group, gains in the intervention group were more pronounced in these traditionally difficult-to treat populations. For patients with chronic depression (37% of all patients), AD use increased by 9.5% in the intervention group and decreased by 8.1% in controls ($P = .03$). For patients with dysthymia (24% of all patients), AD use increased by 17.3% in intervention patients and fell by 6.4% in controls ($P = .028$). For patients with potential personality disorders, AD use increased by 9.9% in intervention patients and fell by 6.0% in controls ($P = .02$). The pattern of improvement of overall depression outcomes (mBDI) for these three difficult-to-treat subgroups was similar to the pattern seen in AD use rates, but did not approach statistical significance.

We examined AD use rates at 6 months in other subgroups for which the intervention might be expected to be more effective, including those with more education, more comorbid disease, more non-AD medications, and those who had received prior mental health treatment. No significant differences between intervention and control groups were found among these groups.

3.6. Evaluation of outcomes hypotheses

As expected, mental health outcomes were better for patients on ADs, regardless of whether they were in the intervention or control group. Those not on ADs at 3 months

Table 4
Rates* of Antidepressant Use by Initial AD Taking Status

	NOT on AD at Enrollment (%)				ON AD at Enrollment (%)			
	N	Intervention (N = 115)	Control (N = 119)	P-value	N	Intervention (N = 117)	Control (N = 110)	P-value
Initial	234	0.0	0.0	N/A	227	100.0	100.0	N/A
3 month	179	29.2	11.0	0.005	181	90.7	87.2	0.50
6 month	176	32.3	10.9	0.001	183	83.4	78.4	0.33

* 3- and 6-month results adjusted for screener covariates: age, gender, mBDI, MHI, and prior AD use.

Table 6
Rates* of antidepressant use at 6 months in traditionally difficult populations not on ADs at enrollment

	N	Intervention (%)	Control (%)	P-value
Chronic depression†	53	42.7	13.9	0.05
Dysthymia	41	47.8	15.6	0.06
Potential Personality Disorder	99	37.1	13.4	0.01

* 3- and 6-month results adjusted for screener covariates: age, gender, mBDI, MHI, and prior AD use.

† Chronic depression is ≥ 4 episodes and > 10 years since onset of first episode.

experienced an average mBDI decrease (improvement) of 3.4 points between screener and 6 months, while those on ADs at 3 months experienced an average mBDI decrease of 6.5 points during the same period ($P = .005$ for the difference regardless of whether they were in the intervention or control group) (data not shown).

Given the benefits of AD use and the increase in AD use for the intervention arm, one can calculate the indirect effect of the intervention on outcomes as the product of improvement in AD use and the impact of that AD use on mBDI outcomes. Both elements of this product were statistically significant. Multiplying the impact on AD use (11.7% from 3 months, Table 2) by the decrease (improvement) in mBDI points associated with using an AD (-3.1 points), we calculated an indirect effect of the pharmacist, because of his or her ability to increase AD use as -0.4 mBDI points ($11.7\% \times -3.1$ points). For those who were not taking ADs initially, the indirect effect of the intervention is -0.6 mBDI points ($18.2\% \times -3.1$ points).

3.7. Trends and contamination

There were 53 primary care providers in the 9 sites included in the study. Although providers ($P = .001$) and sites ($P = .01$) differed significantly in the fraction whose patients took ADs at enrollment, they did not differ in the fraction of patients taking ADs at 3 or 6 months, regardless of enrollment medication status. In addition, there was no interaction between provider and intervention group in predicting AD use at enrollment, 3, or 6 months.

After controlling for site, patients who started the study later (after the median entry time) had a higher chance of being on medications at both the time of the initial screening period ($P = .08$) and after 6 months ($P < .05$). The impact of the experiment was larger (and in fact, was significant earlier in the study) for both AD use and mBDI outcomes.

4. Discussion

The intervention was designed to remove barriers preventing patients both from starting ADs and from staying on ADs. Each of these goals was achieved. The intervention

was most successful for those patients who were not taking an AD at enrollment. The pharmacists were able to initiate and maintain AD use through the 6-month assessment for 32% of the sample, compared to 11% in the control group ($P = .001$). These results corroborate prior findings showing that patients who are “treatment naïve” experience a significant improvement if randomized to an intervention group [1,4,11,13].

The decision to conduct this trial in the real world of clinical practice reduced the intervention’s potential for success. Unlike most other AD studies, we did not automatically exclude any category of depressed patients such as those who were already on ADs, who were unwilling to try ADs, and who had co-morbid conditions such as anxiety or personality disorder. In contrast, clinical trials, for example, which typically report much more robust results (i.e., 50% reduction in BDI score), focus almost entirely on patients without co-morbid conditions or current AD usage. But outcomes in the intervention group were not significantly better than outcomes in the control group. A key reason for this finding was that the intervention itself did not directly affect outcomes. Although the intervention increased the numbers of patients taking ADs, AD use, in turn, did not necessarily improve outcomes. We speculate that in primary care practice many chronically depressed patients with additional co-morbid psychiatric and medical conditions are on ADs despite, at best, incomplete clinical improvement. Even with the intervention significant numbers of patients, whether a result of prior negative experiences or stigma, were still unwilling to try a new course of ADs.

On the other hand, the decrease in BDI points resulting from our intervention were of the same magnitude as those found in the Partners in Care [30,52] studies, which did achieve statistical significance [$n = 1356$; 0.22 effect size in standardized units¹ vs. 0.21 (pharmacist)] [13]. Moreover, our intervention did have potential to improve medication taking for subgroups of patients often considered difficult to treat [8,10,20,25,27] those with dysthymia and chronic recurrent depressive disorder, as well as patients with likely personality disorder (indicated by high NEO scores in our study). In addition, pharmacists were quite successful in increasing AD use for patients who had never tried ADs (or had discontinued them before the study) and for patients who stopped their ADs during the study. Therefore, even though the intervention per se did not lead to significantly better outcomes, adding pharmacists to the primary care patient’s team does have potential to complement other types of treatment.

We also attempted to determine the reasons for these positive results. A regression analysis failed to find a definitive interaction between the intervention and any patient characteristics (e.g., such as sociodemographic variables, co-morbid health conditions or prior mental health care utilization). However, given that we noted a dose-response effect for the pharmacist (based on the amount of time spent with the patient, data not shown), we suspect that interper-

sonal factors—that is, the sustained attention of another health professional—contributed strongly to success. Likewise, both Rost [8] and Wells [52], also using samples considered more representative of community care than prior studies [1,3–5,7,10,11], found that nurse interventions are effective in persuading primary care patients to initiate a new course of AD therapy. Thus, the evidence from all three studies suggests that physician extenders who advocate for patients and help sustain their involvement in their own care can be useful in improving outcomes. For example, a PCP might choose to refer patients reluctant to begin but in need of antidepressants to a pharmacist for further medication education. In addition, other groups with traditionally poorer outcomes, such as those with chronic depression, dysthymia or possible personality disorder, could also be referred to a pharmacist for more aggressive medication monitoring.

The study also suffered from several limitations. Sample size losses in enrollment both increased the ambiguity of the results, and decreased generalizability. The wide confidence intervals for outcomes (e.g., 0–4 mBDI points) cloud simple statistical conclusions and forced reliance on indirect effects analysis. For example, if one were to consider the upper end of the confidence interval for outcomes, the intervention might have achieved a 4 BDI point additional reduction for the sample as a whole and 5 BDI points for the subgroup not on ADs at enrollment. The indirect effects analysis indicates that the statistically nonsignificant result is unlikely because of a lack of power. Study enrollment may have been biased against patients, who did not want to take ADs, but it was not statistically significant, and we have no reason to think it was different from other studies. Second, the control group was not as pure as it might have been because the screening component of the study also drew attention to the depression status of patients in the control group. And the particularly high degree of mental health care utilization in the Boston area meant that many patients in the control group had already received some form of treatment for depression. By reducing the effect size, these two contaminants also appear to have reduced the power to detect an impact of the intervention. Third, both AD use and outcome measures were self-reported (but were validated as part of this study). Fourth, the adequacy of AD dose was not explored explicitly, however, our own results show that the level of continuity was very high, even in the control group, so if a patient was on ADs at 6 months there is an overwhelming chance an adequate dose was received—unlike past studies.

Finally, as the results documented under “Trends and Contamination” suggest, experimental conditions did not remain steady over the course of the study. The benefits in the intervention arm tended to be greater during the first 3 months following implementation at a given site, even though both AD use and outcomes continued to improve over time in the sample as a whole (intervention and control). Three hypotheses, which are by no means mutually exclusive, might account for these findings. Perhaps the trend toward better treatment and follow-up may have continued

during our study. Pharmacists may have set a good example, thus inducing physicians to improve their treatment protocols for their control-group patients. Or, perhaps study participation may have motivated physicians to approach depression more consistently according to AHRQ guidelines.

Our study also highlights a general problem in RCTs that stems from the long lag time between grant application and reported results. In designing and conducting future experiments, researchers need to take into account that underlying practice trends can change quickly. The high rates of patients on ADs at enrollment in our study point to a trend toward better treatment of depression by PCPs (presumably made possible by the increasing acceptability and effectiveness of the SSRIs). Once on an AD, patients in our study stayed on in numbers previously unseen, contrary to previous reports of high discontinuation rates after 3 months [68], decreasing the likelihood of an intervention effect. In other words, the vast literature that continues to report little progress in the age-old problems of under-detection and under-treatment of depression may be outdated.

Thus, our study suggests that policy makers now need to differentiate between two categories of “untreated” depression: the first consists of patients who have not yet been screened and who are likely to benefit from standard treatments and the second consists of patients who are receiving treatment but who are not improving. For primary care patients in the first category, pharmacists (and other extenders to PCPs) remain an underutilized resource who can be instrumental in helping them start and maintain active participation in a course of pharmacotherapy. However, because of a reduction in the stigma toward mental illness and improvements in clinical practice over the past decade, the number of patients in this category has shrunk. As our study shows, many patients screened for depression in primary care practices are already taking AD medications. But unfortunately, some patients with chronic depressive disorders (who visit their PCPs more frequently than those with first or acute episodes) may not improve at all or may still experience some depressive symptoms despite continued AD use. These patients might benefit from more complex treatments, involving still more effective medication regimens and/or psychotherapy. At the same time, it must be acknowledged that the goal of eliminating—rather than significantly reducing—depressive symptomatology may not always be feasible.

Note

¹Difference between baseline and 6 months/SD of scale at baseline.

Acknowledgments

This research was supported by the National Institute of Mental Health under grant RO1 MH56214. We wish to

acknowledge the very important contributions to this project by our clinical pharmacist intervenors: Maryann Kaszuba Pharm.D., Christopher McCoy, Pharm.D., Jill Dischler, Pharm.D., and Anita Wagner, Pharm.D. MPH as well as by Doris Hernandez, our project assistant, and Joshua Kendall for his editorial assistance.

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